EVERY QCQS ALGEBRAIC SPACE ADMITS A FINITE FILTRATION BY CLOSED SUBSPACES WHOSE STRATA BECOME RESOLUTION SPACES AFTER ADMISSIBLE BLOWING UPS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13880-13892 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 13894-13933, the complete author proof (38 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The proof's first paragraph marks pullback details omitted; the induction core is written in full inside the retained span.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Absolute Noetherian approximation for qcqs spaces (Limits of Spaces, proposition-approximate), ascent of the resolution property along affine morphisms (Derived Categories of Spaces, lemma-resolution-property-goes-up-affine), and the single-step blowup lemma of the same chapter (lemma-blow-up-resolution-property).

Source reference. Original lemma 0GV5: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13833–13839. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03I5: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13933–13933. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07SU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13933–13933. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GUU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13933–13933. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 03BS: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13933–13933. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. There exists a $t \geq 0$ and closed subspaces*

$$X \supset Z_0 \supset Z_1 \supset \ldots \supset Z_t = \emptyset$$

such that $Z_i \to X$ is of finite presentation, $Z_0 \subset X$ is a thickening, and for each $i = 0, \ldots t-1$ there exists a $(Z_i \setminus Z_{i-1})$-admissible blowing up $Z_i' \to Z_i$ such that Z_i' has the resolution property.

Proof. In this paragraph we use absolute Noetherian approximation to reduce to the case of algebraic spaces of finite presentation over $\mathrm{Spec}(\mathbf{Z})$. We may view X as an algebraic space over $\mathrm{Spec}(\mathbf{Z})$, see Spaces, Definition 03I5 and Properties of Spaces, Definition 03BS. Thus we may apply Limits of Spaces, Proposition 07SU. It follows that we can find an affine morphism $X \to X_0$ with X_0 of finite presentation over $\mathbf{Z}$. If we can prove the lemma for X_0, then we can pull back the stratification and the centers of the blowing ups to X and get the result for X; this uses that the resolution property goes up along affine morphisms (Derived Categories of Spaces, Lemma 0GUU) and that the strict transform of an affine morphism is affine – details omitted. This reduces us to the case discussed in the next paragraph.

Assume X is of finite presentation over $\mathbf{Z}$. Then X is Noetherian and $|X|$ is a Noetherian topological space (with finitely many irreducible components) of finite dimension. Hence we may use induction on $\dim(|X|)$. By Lemma 0GV5 there exists a dense open $U \subset X$ and a U-admissible blowing up $X' \to X$ such that X' has the resolution property. Set $Z_0 = X$ and let $Z_1 \subset X$ be the reduced closed subspace with $|Z_1| = |X| \setminus |U|$. By induction we find an integer $t \geq 0$ and a filtration

$$Z_1 \supset Z_{1,0} \supset Z_{1,1} \supset \ldots \supset Z_{1,t} = \emptyset$$

by closed subspaces, where $Z_{1,0} \to Z_1$ is a thickening and there exist $(Z_{1,i} \setminus Z_{1,i+1})$-admissible blowing ups $Z_{1,i}' \to Z_{1,i}$ such that $Z_{1,i}'$ has the resolution property. Since Z_1 is reduced, we have $Z_1 = Z_{1,0}$. Hence we can set $Z_i = Z_{1,i-1}$ and $Z_i' = Z_{1,i-1}'$ for $i \geq 1$ and the lemma is proved. □